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BIO 301 Final terms Subjective

1. Define epigenetic? 2 marks

Epigenetics literally means 'above' the genetics. Epigenetic trait is a stable and heritable phenotype resulting from changes in chromosome without alterations in the DNA sequence.

2. Complication of inheritance? 3 marks

Multiple genes - epistasis, polygenic traits

Genes & the environment - sexinfluenced traits, incomplete penetrance

A common recessive inheritnace can give dominant pattern

Autosomal dominant inheritance with nonpenetrance

Autosomal dominant inheritance with variable expression.

3. Mitochondrial disorders? 3 marks

Myoclonic epilepsy

Mitochondrial recessive ataxia

Leber hereditary optic neuropathy

Sensory ataxia neuropathy

4. Cause of epigenetic? 3 marks

CAUSES OF EPIGENETICS:

1. DNA Methylation

2. Histone Modification

3. miRNAs

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5. Point mutation? 2 Marks

Point mutations affect single sites on DNA. Substitution of one base for another.

Deletion/addition of a single base or small number of bases.

6. Point mutation and how mutation changes in the chromosomes structure? 5 marks

Point mutations affect single sites on DNA. Substitution of one base for another.

Deletion/addition of a single base or small number of bases.

VARIATION IN CHROMOSOME STRUCTURE

Deletions

Translocations

Duplications

Inversions

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7. How mutations alter gene function? 10 marks

SPLICE SITE MUTATIONS HAVE SEVERE EFFECTS

Splice sites site mutations usually have severe effects on protein functions. Often this kind of mutation produce one or more exon skipping or larger fragments of DNA.

EFFECTS OF NULL ALLELE AND HYPOMORPH

Null allele results in loss of function. Usually no gene product. Hypomorph alleles□results in decreased activity of the proteins.

EFFECTS OF MUTATION ON GENE FUNCTION

Gain of function - increased activity. Dominant negativeantagonizes normal product. Neomorph - novel activity.

Missense mutations have sometimes severe effects on protein function while some times less severe effects. It depends on the type of amino acid which was replaced.

8. Epigenetic modification? 10 marks

EPIGENETIC MODIFICATIONS Cytosine methylation

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Histones modification

Non-coding RNAs

(siRNAs)

All above regulate gene expression.

CYTOSINE METHYLATION OCCURS AT UPSTREAM REGIONS OF GENES

Often located just upstream of genes (promoter regions). Associated with attenuation of expression of nearby genes.

HISTONES MODIFICATIONS

Histones are the proteins that organize the genetic material. Have a high percentage of basic amino acids, which gives histones an overall positive charge, while DNA is negative charged.

ACETYLATED HISTONES-GENES ARE ACTIVE

When histones are acetylated, chromatin is open and genes are potentially active. When histones not tagged, deacetylated, the chromatin condenses and genes silenced.

HISTONES MODIFICATIONS:

Most histones modifications occurs on the extended tails of histone proteins.

HISTONES ACETYLATION

Two enzyme types involved in histone acetylation. HAT: Histone acetyltransferase HDAC: Histone deacetylase.

EPIGENETIC MODIFICATION - NON CODING RNA

A new mechanism for gene regulation. RNA which is not used for making proteins can be cleaved and used to inhibit protein-coding RNAs (siRNAs, microRNAs).

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9. Chromosomal aberration 10 marks?

ABERRATIONS : Any change in normal number or structure of chromosomes. Affects individuals by altering the amounts of products of the genes involved. Altered amounts may cause anomalies directly or may alter the balance of genes.

CLASSIFICATION:

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Numerical (usually due to de novo error in meiosis).

Structural (may be due to de novo error in meiosis or inherited).

Different cell lines (occurs postzygotically).

NUMERICAL CHANGES

1. Aneuploidy

Hypoploidy

Hyperploidy

2. Euploidy

Monoploidy

Diploidy

Polyploidy

STRUCTURAL

Translocations

Deletions

Duplications

Inversions

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